Dr. Ludwig M. Böss

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Education & Academic Positions

Jan 2025–now
Chicago, IL
Postdoctoral Researcher, The University of Chicago

Mar 2024–Dec 2024
Munich
Postdoctoral Researcher, Ludwig-Maximilians-Universität

Apr 2020–Mar 2024
Munich
Astrophysics, PhD, Ludwig-Maximilians-Universität, *magna cum laude*

Thesis: On Simulations of Cosmic Rays in the Large-Scale Structure of the Universe
Supervisors: Prof. Dr. Harald Lesch & Prof. Dr. Klaus Dolag

Sep 2017–Apr 2020
Munich
Astrophysics, M.Sc., Ludwig-Maximilians-Universität

Thesis: Cosmic Rays in Galaxy Clusters - An on-the-fly Fokker-Planck Solver for OpenGadget3; Supervisor: Prof. Dr. Klaus Dolag

Sep 2014–Aug 2017
Munich
Physik, B.Sc., Ludwig-Maximilians-Universität

Thesis: Radial Orbit Instability - Analysis of geometry in unperturbed and perturbed systems; Supervisor: Prof. Dr. Andreas Burkert

Sep 2011–Aug 2014
Munich
Musicology, B.A., Ludwig-Maximilians-Universität

Thesis: Witold Lutoslawski's Concerto for Orchestra in the context of Socialist Realism; Supervisor: Prof. Dr. Wolfgang Rathert

Conferences and Workshops (Selected)

July 2026 **ICM Theory and Computation Workshop**, Brno, *CZ (Invited Talk)*

September 2025 **Relativistic Plasma Physics: From the Lab to the Cosmos**, Santa Barbara, *USA*

January 2025 **Particle Acceleration and Transport: from the Sun to Extragalactic Sources**, Rende, *IT (Invited Talk)*

September 2024 **Galaxy Clusters & Radio Relics II**, CfA, Cambridge, *USA (Invited Talk)*

June 2024 **ICM Theory and Computation Workshop**, Ann Arbor, *USA (Invited Talk)*

Invited Seminar Talks (Selected)

November 2026 **KIPAC Seminar**, Stanford, *USA*

February 2026 **UW-Madison Astronomy Colloquium**, Madison, *USA*

December 2025 **Twin Talks - PPPL Seminar**, Princeton, *USA*

November 2025 **Michigan State University Astronomy Seminar**, East Lansing, *USA*

October 2025 **THEA Group Seminar**, New York, *USA*

Awarded Computing Time

2025-2026 **ALCF Director's Discretionary Allocation (PI)**, Cosmic Rays in Cosmological Simulations - Bridging Scales
60,000 Node hours, Aurora

2024 **GCS Project (PI)**, Simulations of Cosmic Rays in Galaxy Clusters at Unprecedented Resolution
29 Million CPUh, SuperMUC-NG

2023 **GAUSS Large Scale Project (co-I, PI: Klaus Dolag)**, The Local Universe: Galaxies, Clusters, The LSS and Cosmic Rays
69 Million CPUh, SuperMUC-NG

2022 **C2PAP Computing Grant (PI)**, Galaxy Clusters with Spectral Cosmic Rays
6.5 Million CPUh, SuperMUC-NG

2020 **GAUSS Large Scale Project (co-I, PI: Klaus Dolag)**, COMPASS
50 Million CPUh, SuperMUC-NG

Awarded Observation Time

2025 **XRISM GO Cycle 2 (Co-PI with Irina Zhuravleva)**, ICM Turbulence in the Merging Chaos of Abell 2256
2 pointings, 290ks total

Submitted Proposals

2026 **INCITE 2027 (PI)**, Cosmic Ray Acceleration at Astrophysical Shocks
2.29 Million Node hours, Aurora

2026 **INCITE 2026 (renewal, co-I, PI: Sasha Philippov)**, Extreme Plasmas at the Spacetime Frontier: Kinetic effects in Black hole Accretion
937,600 Node hours, Frontier

Student Supervision

2025-now **PhD Thesis**, Student: Anna Ivleva

Assisting supervisor

2024-now **PhD Thesis**, Student: Daniel Karner

Assisting supervisor

2023 **Masters Thesis**, Students: Daniel Karner, Julian Sommer

Assisting supervisor

Skills

Languages

German	First language
English	Fluent
French	Conversation skills

Programming skills

Julia	●●●●○	C/C++	●●●●○	Python	●●●●○	Fortran	●●●○○
SQL	●●●●○	HTML/CSS	●●○○○				

First-Author Publications

in prep — **Entity – Hardware-agnostic Particle-in-Cell Code for Plasma Astrophysics. IV: Hybrid-PIC Module**, **Böss, L.**, Orusa, L., Galishnikova, A., Hakobyan, H., Caprioli, D.

in prep — **Whistler-suppressed thermal conduction and diamagnetic heat flux in the intra-cluster medium**, **Böss, L.**, Palchoudhury, P., Ruszkowski, M., Caprioli, D.

submitted — **A shock finding algorithm for meshless-finite-mass hydrodynamic schemes**, **Böss, L.**, Groth, F., Dolag, K., Caprioli, D., arXiv:2610.38700

accepted — **Entity – Hardware-agnostic Particle-in-Cell Code for Plasma Astrophysics. III: Higher-order shape functions & generalized field stencils**, **Böss, L.**, Vanthieghem, A., Hakobyan, H., Gorbunov, E., Caprioli, D., arXiv:2605.15260

2026 — **Simulating the LOcal Web (SLOW) - *VI*: γ-ray Emission from the Local Universe**, **Böss, L.**, Khabibullin, I., Karner, D., Dolag, K., Steinwandel, U., Hernández-Martínez, E., Sorce, J. G., A&A, 713, id.A43.

2024 — **Simulating the LOcal Web (SLOW) - *III*: Synchrotron Emission from the Local Cosmic Web**, **Böss, L.**, Dolag, K., Steinwandel, U., Hernández-Martínez, E., Khabibullin, I., Seidel, B., Sorce, J. G., A&A, 692, id.A232, 25 pp.

2023 — **A formation mechanism for 'Wrong Way' Radio Relics**, **Böss, L.**, Steinwandel, U., Dolag, K, ApJL, 957, L16

2023 — **CRESCENDO: An on-the-fly Fokker-Planck Solver for Spectral Cosmic Rays in Cosmological Simulations**, **Böss, L.**, Steinwandel, U., Dolag, K., Lesch, H., MNRAS, 519,1, pp.548-572

Co-Author Publications

submitted — **The OpenGadget3 Code for Cosmological Simulations**, Dolag K., Karademir G. S., Tornatore L., Ragagnin A., Valentini M., **Böss L.**, Groth F., et al., 2026, arXiv:2609.17658

Contributions: Code development, scientific input.

submitted — **CRESCENDO II: Spectral cosmic rays with improved energy losses and realistic supernova seeding**, Karner D., **Böss L.**, Dolag K., Khabibullin I., arXiv:2607.14363

Contributions: Code development, scientific input, student supervision.

accepted — **Simulating the LOcal Web (SLOW): VII. Intergalactic magnetic field models for multi-messenger applications**, Stoiber J., Dolag K., Capel F., Seidel B., Kachelrieß M., **Böss L.**, Sorce J. G., arXiv:2607.06665

Contributions: Simulation data, scientific input

2026 — **OpenGadget3 GPU solver tests**, Ragagnin A., Karademir G. S., Groth F., Dolag K., **Böss L.**, Castro T., Hariharan N., et al., Astronomy and Computing, Volume 57, id.101131

Contributions: Code development, testing.

2026 — **Entity – Hardware-agnostic Particle-in-Cell Code for Plasma Astrophysics. I: Curvilinear Special Relativistic Module**, Hakobyan, H., **Böss, L.**, Cai, Y., Chernoglazov, A., Galishnikova, A., Gorbunov, E., et al., ApJS, 285, 1, id.11

Contributions: Code development, scaling tests on Aurora, shock setup and accompanying section in paper.

2026 **Shock Acceleration in the Intracluster Medium: Implications of Micromirror Confinement**, Diesing, R., **Böss, L.**, Caprioli, D., ApJ, 999, 1, id.54

Contributions: Underlying simulation and interpretation of the results.

2026 **Cosmological Zoom-In Simulation of Odd Radio Circles as Merger-Driven Shocks in Galaxy Groups**, Ivleva, Anna, **Böss, L.**, Dolag, Klaus, Koribalski, B. S., Khabibullin, I., et al. , A&A 706, A80.

Contributions: Student supervision, simulation setup, help with data analysis.

2026 **Magnetic Fields in the Shapley Supercluster Core with POSSUM: Challenging Model Predictions**, Alonso-López, D., O'Sullivan, S., Bonafede, A., **Böss, L.**, et al., A&A, 705, A143

Contributions: Simulation data, help with interpretation of the results.

2026 **Two intersecting radio shells: relics of galaxy merger shocks?**, Koribalski, B. S., Dolag, K., Khabibullin, I., **Böss, L.**, Ivleva, A., Norris, R., PASA, 43, id.e061

Contributions: Scientific input.

2026 **Single- and double-headed odd radio circles in the LOFAR surveys**, De Gasperin, F., Brüggen, M., ... , **Böss, L.**, et al. , A&A 709, A154.

Contributions: Scientific input.

2024 **The glow of axion quark nugget dark matter: II. Galaxy clusters**, Sommer, J. S., Dolag, K., **Böss, L.**, Khabibullin, I., Liang, X., et al., A&A, 691, A38.

Contributions: Scientific input.

2024 **Towards cosmological simulations of the magnetized intracluster medium with resolved Coulomb collision scale**, Steinwandel, U., Dolag, K., **Böss, L.**, Marin-Gilabert, T., ApJ, 967, 2, id.125, 19 pp

Contributions: Scientific input, Fig. 3, 4, 10 & 13 in publication.

2023 **Insights on the origin of ORCs from cosmological simulations**, Dolag, K., **Böss, L.**, Koribalski, B., Steinwandel, U., Valentini, M., ApJ, 945, 74

Contributions: Modeling of the CR component, related figures and chapters in the paper. Interactive figure in the online publication.

2022 **Virgo: Scalable Unsupervised Classification of Cosmological Shock Waves**, Lamparth, M., **Böss, L.**, Steinwandel, U., Dolag, K., arXiv:2208.06859

Contributions: Scientific input, development contribution, all figures in publication.

2022 **On the small scale turbulent dynamo in the intra cluster medium: A comparison to dynamo theory**, Steinwandel, U., **Böss, L.**, Dolag, K., Lesch, H., ApJ, 933, 2, 131

Contributions: Analysis tools and scripts. Related test simulations.

2019 **WVTICs - SPH initial conditions for everyone**, Arth, A., Donnert, J, Steinwandel, U., **Böss, L.**, et al., arXiv:1907.11250

Contributions: Implementation of the artificial bias correction and writing the corresponding section. Rerunning the tests and figures for the paper.